



Fig. 3: Battery used in the radio

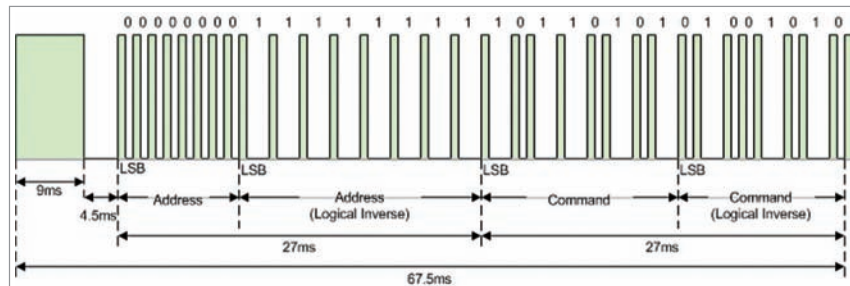


Fig. 4: NEC message frame with timing

TABLE 1
BILL OF MATERIALS

S.No.	Description	Reference	Quantity
1	220μF/16V electrolytic capacitor	C7, C8, C9	3
2	470μF/16V electrolytic capacitor	C6, C10	2
3	100nF capacitor	C1, C2, C4	3
4	22nF capacitor	C3	1
5	LD1117V33 (TO-220) or AMS1117-3.3 (SOT-223) 3.3 volts fixed voltage regulator	IC1	1
6	ESP-01 or ESP-01S	U1	1
7	RDA5807M FM stereo module RRD-102 V2	U2	1
8	4 pin 2.54mm header	LCD/OLED	1
9	3.5mm stereo audio connector	X2	1
10	3.7V/500mAh lithium-ion battery		1
11	TP4056 charger board	X1	1
12	Micro USB cable		1
13	Earphone or headphone, 32 ohms		1
14	SWG#22 wire 75cm length for antenna		1
15	23mm 128x32 OLED display SSD1306		1
16	38kHz IR 3.3V remote receiver and IR remote transmitter	U3	1

IR remote control along with RDS display using a I2C display (LCD or OLED).

Banking on these advantages we can quickly develop and integrate an FM radio with shortest development time using ESP-01. Fig. 1 shows the author's prototype of stereo FM

(U2), 38kHz IR remote receiver 3.3V module (U3), 23mm (0.91-inch) 128x32 OLED display SSD1306, 32-ohm earphones or headphones, and a few other components.

Both the ESP-01 and the IR receiver require a 3.3V supply. You can use a 1117 regulator with a fixed

radio with RDS using ESP-01.

To get started, let us look at the bill of materials in Table 1.

Circuit and working

The circuit diagram for the remote-controlled stereo FM radio with RDS using the ESP-01 is depicted in Fig. 2. It comprises the 3.3V voltage regulator LD117AV33 (IC1), ESP-01 (U1), RDA5807M FM stereo module RRD-102 V2

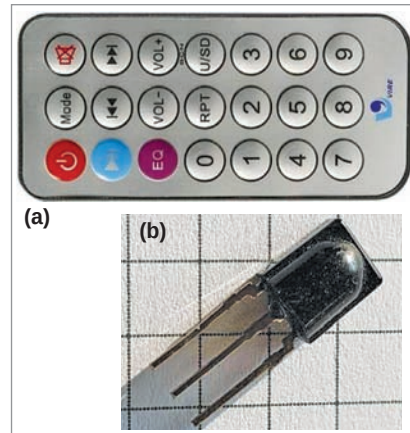


Fig. 5: (a) IR remote, and (b) IR receiver module

3.3V output in a TO-220 package or opt for a 3.3V breakout board utilising the AMS1117-3.3 in an SOT-223 package. To ensure smooth operation, bulk and bypass capacitors are added to both the input and output of the 3.3V regulator. The circuit was tested with both TO-220 and SOT-223 package 3.3V voltage regulators.

For home installations, a 5V power supply from a 230V AC to 5V DC converter can be employed. Alternatively, you can replace the USB connector (X1) with a 3.7V lithium-ion cell for a lighter, portable setup, as indicated in Fig. 3. The TP4056 charger board is utilised to charge the battery.

Software

Compile the code using Arduino 1.8.19 IDE. To begin, install the IDE and ESP boards. You will also need libraries such as Radio-2.0.0, Irremote-3.9.0, SSD1306Ascii-1.3.5, RDSParser, and RDA5807M. These libraries can be installed via the Arduino library manager found in the 'Sketch' menu bar. The SSD1306Ascii-1.3.5 library is used for the OLED display, indicating the FM channel frequency, radio volume level, and RDS message information.

The RDS message display utilises the Radio-2.0.0 library, while the NEC receiver uses Irremote-3.9.0 for